

TRUNKBELAY · PRECIS

TrunkBelay design precis

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DOCUMENT CONTROL

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REVISION HISTORY

Version	Date	Author	Change
0.3	2026-10-03	Amish Chadha	TRL 3; constructable design of TKB-DDR-002 and figures from TKB-CAL-001
0.2	2026-10-03	Amish Chadha	TRL 2 precis; how it works, components with BOM numbers, key design choices, first-order numbers, safety
0.1	2026-09-30	Amish Chadha	Initial scaffold

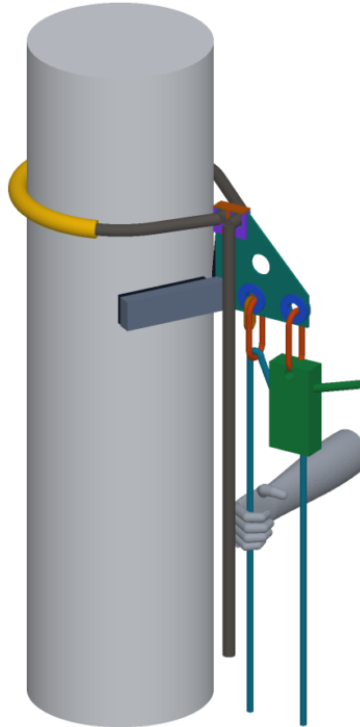
TrunkBelay design precis

Lets a fellow climber anchor and lower a climber stranded high on a palm trunk.

TrunkBelay

github.com/BoujeeEnjinia1701/trunkbelay

CONCEPT, NOT FOR FABRICATION



Seen from the front right and above, 24 deg elevation. Grey: Palm trunk, 300 mm (site), Rescuer's hand on the brake strand (scale) for scale

Figure 1. The collar on a 300 mm trunk, seen from the front right and above. The grey hand on the brake strand gives the scale.

■ How it works

TrunkBelay is a steel collar that grips a palm trunk harder the more load hangs from it, with a commodity rope descender clipped to it. The rescuer climbs on their own climbing device to just below the stranded climber and works in four stages.

1. **Fix the collar.** The rescuer holds the collar frame against the trunk, with its two rubber-faced bearing bars touching the bark, passes the chain round the trunk, drops a link into the open slot on top of the frame, pulls the chain snug and slips the keeper and lock pin on. The frame is set at least 300 mm above the point where the rope will meet the victim.
2. **Rig the lowering.** The descender is clipped to the load eye at the outer end of the frame and the rope reeved through it. The brake strand turns over a carabiner on the spare eye and runs down to the rope bag, which is lowered to the helpers on the ground.

3. **Take the weight.** The rescuer fits an evacuation triangle round the victim's hips and a chest sling round the chest, clips both with the rope's end loop into the victim carabiner, and pulls all slack through the descender, which locks.
4. **Free and lower.** The rescuer frees the victim's feet from the climbing device. The victim turns head up in the triangle and hangs from the collar. The rescuer opens the descender with its handle and lowers the victim to the helpers, holding the brake strand. The rescuer then takes the kit down and descends on their own device.

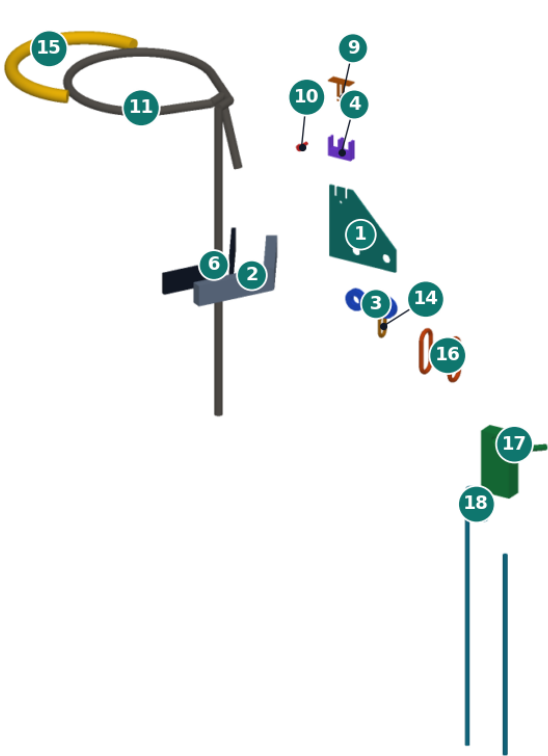
The load eye sits 150 mm out from the frame. The pull of the rope on the eye tilts the frame on the trunk: the bearing bars press the bark low on the near side and the chain presses it high on the far side. The friction at those two places rises in step with the load, so the collar is self-locking, like a drawer that jams when it is pulled from one corner (TKB-CAL-001, section B).

TrunkBelay: exploded view

github.com/BoujeeEnjinia1701/trunkbelay

CONCEPT, NOT FOR FABRICATION

- 1 Spine plate
- 2 Bearing bars (2)
- 3 Eye doubler rings (4)
- 4 Cleat cheeks (2)
- 6 Rubber bark pads (2)
- 9 Keeper
- 10 Lock pin
- 11 Chain, grade 80, 6 mm (envelope)
- 14 Tether quick link
- 15 Chain sleeve
- 16 Carabiners (load, brake)
- 17 Auto-locking descender
- 18 Lowering rope, 11 mm



Seen from the front right and above, 24 deg elevation; numbers match bom/bom.csv

Figure 2. Exploded view; numbers match bom/bom.csv.

■ Components

Table 1. Components and their BOM lines.

BOM	Component	Role
1	Spine plate	5 mm steel plate in the frame's centre plane: carries the bearing bars at its foot, the chain slots at its top and the load and spare eyes at its outer end
2	Bearing bars (2)	50 x 25 mm steel box section set in a 120 deg V at the foot of the spine; they bear on the bark through the pads
3	Eye doubler rings (4)	Thicken both eyes to 13 mm so a carabiner bears on a rounded, wide edge
4	Cleat cheeks (2)	Stiffen the slotted top of the spine; notched so the chain links bear on the 5 mm web
5	Welding and galvanising	The frame is one welded, hot-dip galvanised piece
6, 7, 8	Rubber bark pads (2), screws, adhesive	10 mm rubber on each bar spreads the load on the bark and adds grip
9, 10	Keeper and lock pin	Hold both chain links down in their slots
11 to 14	Chain set	6 mm grade 80 chain, 110 links, with a master link on the adjustable end and a quick link tethering the fixed end to the spare eye
15	Chain sleeve	600 mm of tubular webbing on the chain where it bears on the back of the trunk
16	Carabiners (3)	Load eye to descender, brake redirect, victim connector
17	Auto-locking descender	Controls the lowering; locks if the rescuer lets go
18	Lowering rope	30 m of 11 mm semi-static rope
19, 20	Evacuation triangle and chest sling	Hold the victim head up once the feet are freed
21	Rope and carry bag	Carries the kit up the trunk; lowered to the ground with the spare rope

■ Key design choices

All were made under Amish's pre-approval of 2026-10-03 and are recorded in TKB-DDR-001 and TKB-DDR-002, except where a requirement is not met: those are open in the design decisions register (TKB-DEC-001).

- **Self-locking frame, not a tightened band.** The grip comes from the load's lever on the frame, so it does not depend on how hard the rescuer pulled the chain. The chain only has to be snug.
- **A V of two bars, not a curved saddle.** A 120 deg V touches any trunk from 200 to 450 mm on two lines and centres itself; a curved saddle fits only one size.
- **A generic slotted cleat, no ratchet.** Chain links drop into two plain slots in the spine top and are held by a keeper and pin. This keeps clear of the ratcheting cinch geometry of the 3M DBI-SALA Cynch-Lok (the patent design-around) and has no moving parts to clog with bark.
- **Auto-locking descender.** The conservative choice: if the rescuer is hit, faints or lets go, the descender holds the victim.
- **Brake strand to the ground.** The spare rope goes down to the helpers, so the rescuer does not manage a rope bag at height.
- **Steel, welded and galvanised.** Every part can be cut, drilled and welded in a village fabrication shop; aluminium is lighter and is one of the options for the kit mass (TKB-DEC-001).

■ First-order numbers

From TKB-CAL-001; assumptions are stated there.

Table 2. Key figures (estimates).

Quantity	Value
Trunk range	200 to 450 mm across, no tools
Working load on the collar, 100 kg person	0.99 kN
Static load the collar is sized for (R1)	2.5 kN
Friction needed to lock, 200 mm trunk (worst)	0.58, against about 0.9 assumed on wet bark; locking factor 1.54
Frame settles under 2.5 kN	About 10 mm
Strength at 2.5 kN	Bars 2.1 times yield; chain 8.1 times its breaking force; rope 5.7 times at the knot
Hand force on the brake strand, 100 kg	About 80 N
Rope	30 m; 28 m needed for a 25 m lowering
Rigging time once in position	About 4.5 minutes (estimate)
Kit mass	8.9 kg (collar 4.4 kg)
Cost	Value-engineering target: USD 900. Estimated cost of the constructable design: USD 758 (USD 142 under the target)

■ Patent design-arounds

From the preliminary patent, trademark and prior-art screen (not legal advice):

- Generic chain-and-cleat collar; avoid the 3M Cynch-Lok ratcheting cinch geometry and cleated-plate arrangement. The constructable design has no ratchet, no cam and no cleated gripping plate: plain slots, rubber-faced bars and a lever.
- CA2681870C (Bashlin) lapsed in 2015; WO2012131439A1 ceased; neither blocks the concept.

■ Shared blocks

- CalRig proof-load (collar slip and proof tests on trunk sections)
- PalmSpan collar data (shelved sibling), for grip across palm sizes

■ Safety

Safety: TrunkBelay is rope rescue and work-at-height equipment. A slip of the collar, a wrongly fitted triangle or a lost brake hand can drop a person from 6 to 25 m. It is published as an open engineering reference, not certified rescue or fall-protection equipment.

- Only people trained in the drill use it, and they practise at low height first.
- Always call emergency services as well; TrunkBelay is for the minutes before they arrive.
- The collar is not used on a person before it has been proof-tested at 2.5 kN on cut trunk sections and the drill has been run with a dummy at low height (TRL 4). The build plan's safety stops gate both.
- The collar is loaded gently. All slack is taken in before the victim's feet are freed, so the victim never drops onto the collar (TKB-CAL-001, A2).
- The collar is set above the point where the rope meets the victim, so it is never pulled upward.
- A person hanging head down for a long time needs medical assessment after rescue.

This design is published as an open engineering reference. It is not certified equipment.